F12 4th CAR

Lecture 31: Section 4.8
Applications and Models

Solving right triangles

Navigational bearings

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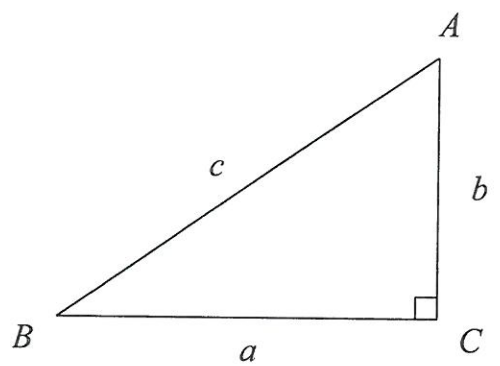
Solving Right Triangles

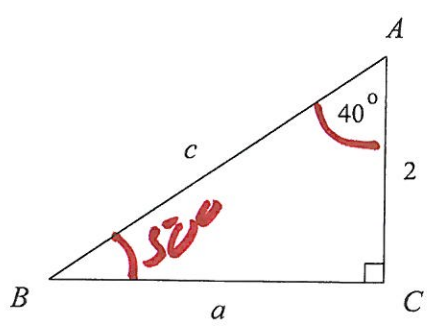
To solve a right triangle means to find the missing lengths of its sides and measurements of its angles.



$$a^2 + b^2 = c^2 \; ; \; A + B = 90^\circ$$

<u>ex.</u> Solve the triangle. $B = 90^\circ - 40^\circ = 50^\circ$

1)



$\frac{opp}{adj} = \tan 40^\circ = \frac{a}{2} \rightarrow \boxed{a = 2\tan 40^\circ}$

calc $\rightarrow \approx 1.68$

$\frac{adj}{hyp} = \cos 40^\circ = \frac{2}{c} \rightarrow \boxed{c = \frac{2}{\cos 40^\circ}}$

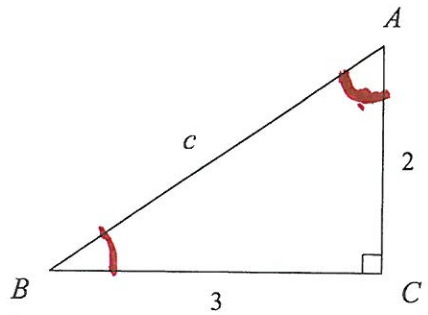
calc ≈ 2.61

$c^2 = 2^2 + 3^2$

$c^2 = 13$

$\boxed{c = \sqrt{13}}$

2)



$\frac{opp}{adj} = \tan A = \frac{3}{2} \rightarrow \boxed{A = \tan^{-1}\left(\frac{3}{2}\right)}$

$\approx 56°$

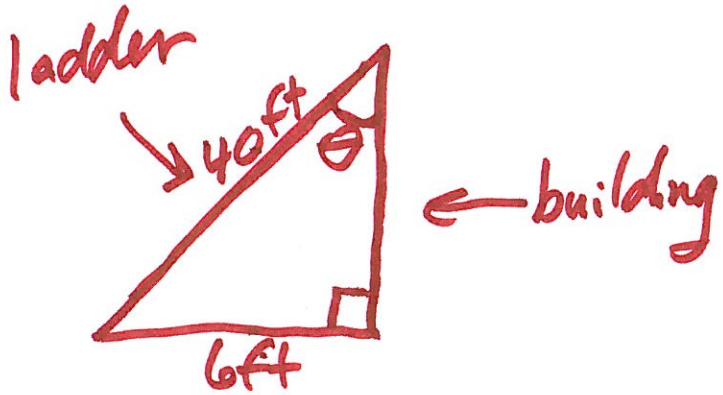
$\tan B = \frac{2}{3} \rightarrow \boxed{B = \tan^{-1}\left(\frac{2}{3}\right)}$

or

$\boxed{B = \arctan\left(\frac{2}{3}\right)}$

calc $\approx 34°$

ex. A 40-ft ladder leans against a building. If the base of the ladder is 6 ft from the base of the building. What is the angle formed by the ladder and the building?

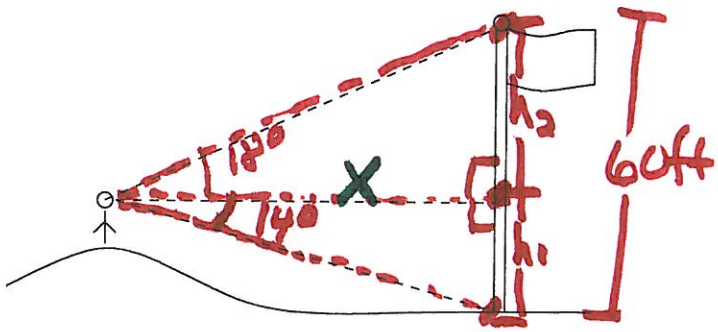


$\frac{opp}{hyp} = \sin\theta = \frac{6}{40}$

$\boxed{\theta = \arcsin\left(\frac{6}{40}\right)}$

calc ≈ 8.6[redacted]

ex. A woman standing on a hill sees a flagpole that she knows is 60 ft tall. The angle of depression to the bottom of the pole is 14°, and the angle of elevation to the top of the pole is 18°. Find her distance x to the pole.



$h_1 + h_2 = 60\text{ft}$

bottom Δ

$\tan 14^\circ = \frac{h_1}{x} \rightarrow h_1 = x \tan 14^\circ$

top Δ

$\tan 18^\circ = \frac{h_2}{x} \rightarrow h_2 = x \tan 18^\circ$

$h_1 + h_2 = 60 \rightarrow x \tan 14^\circ + x \tan 18^\circ = 60$

$x(\tan 14^\circ + \tan 18^\circ) = 60$

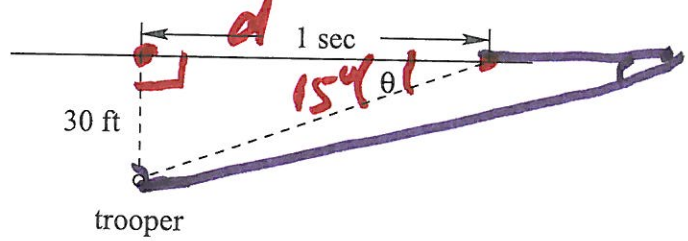
$$\boxed{x = \frac{60}{\tan 14^\circ + \tan 18^\circ}} \rightarrow \text{calc} \approx 104\text{ft.}$$

1) Let d = distance traveled in 1 sec.

Use tan to find d in ft/sec.

$\tan 15° = \frac{30\text{ft}}{d} \rightarrow d = \frac{30\text{ft}}{\tan 15°} \rightarrow$ calc $d \approx 111.96\text{ft/sec}$

ex. A state trooper is hidden 30 ft from a highway. One second after a truck passes, the angle between the highway and the line of observation from the patrol car to the truck is measured.



1) If the angle measures 15°, how fast is the truck traveling? Express the answer in feet per second and in miles per hour.

convert to

$$\frac{111.96\,\cancel{\text{ft}}}{1\,\cancel{\text{sec}}} \cdot \frac{3600\,\cancel{\text{sec}}}{1\text{hr}} \cdot \frac{1\text{mi}}{5280\,\cancel{\text{ft}}} = \boxed{\frac{111.96(3600)}{5280}\ \frac{\text{miles}}{\text{hour}}}$$

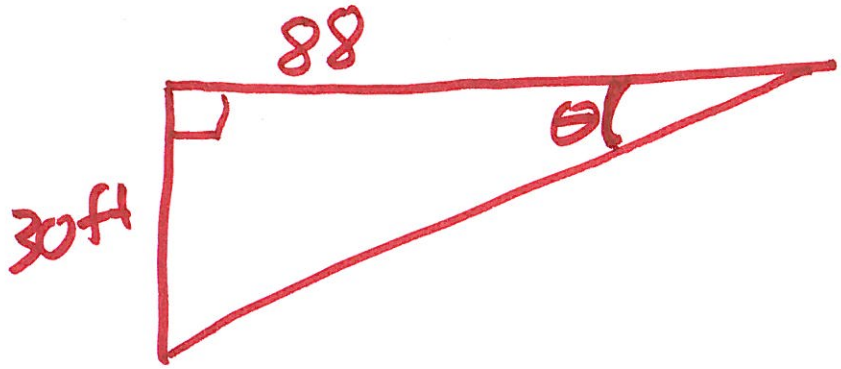
$$\approx \frac{76.34\text{ mi}}{\text{hr}}$$

① Need to know the # of feet traveled in 1 sec. if going 60 mph.

2) If the speed limit is 55 miles per hour and a speeding ticket is issued for speeds of 5 miles per hour or more over the limit, for what angles should the trooper issue a ticket?

$$\frac{60\,\cancel{\text{mi}}}{1\,\cancel{\text{hr}}} \cdot \frac{5280\text{ft}}{1\,\cancel{\text{mi}}} \cdot \frac{1\,\cancel{\text{hr}}}{3600\text{sec}} = \frac{60(5280)}{3600} = 88\ \frac{\text{ft}}{\text{sec}}$$

② Then find the angle associated with the triangle.

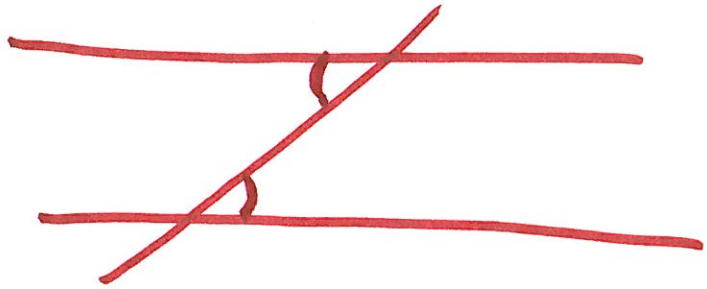


$$\tan\theta = \frac{30}{88}$$

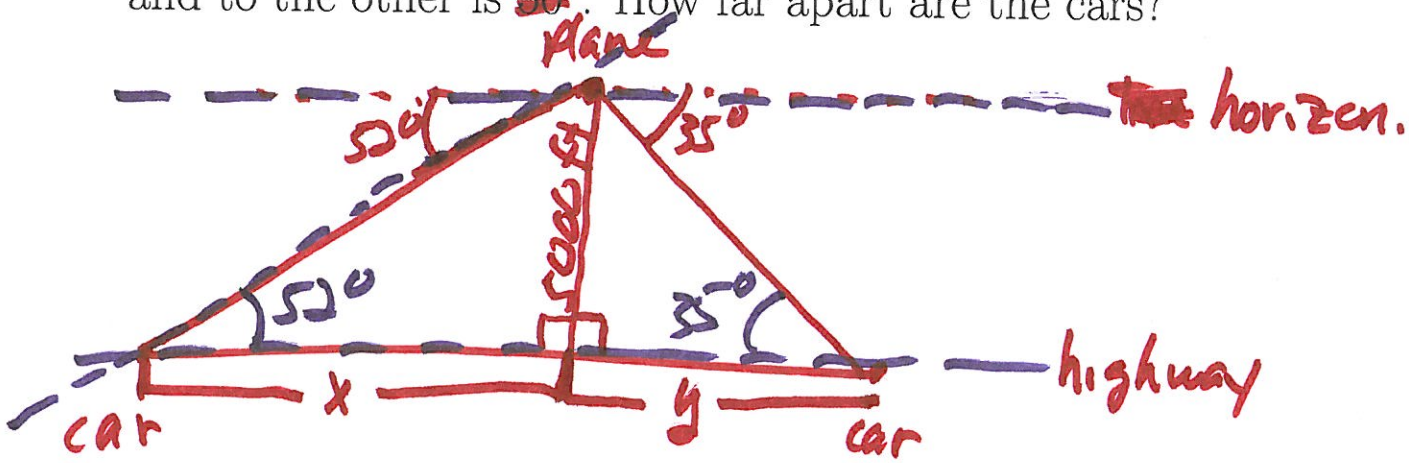
$$\theta = \tan^{-1}\left(\frac{30}{88}\right)$$

$$\boxed{\text{so } \theta < \tan^{-1}\left(\frac{30}{88}\right)}$$

calc $\theta < 18.85^\circ$



ex. An airplane is flying at an elevation of 5000 ft, directly above a straight highway. Two motorists are driving cars on the highway on opposite sides of the plane, and the angle of depression to one car is 35° and to the other is ~~50°~~ 52°. How far apart are the cars?



Find x+y.

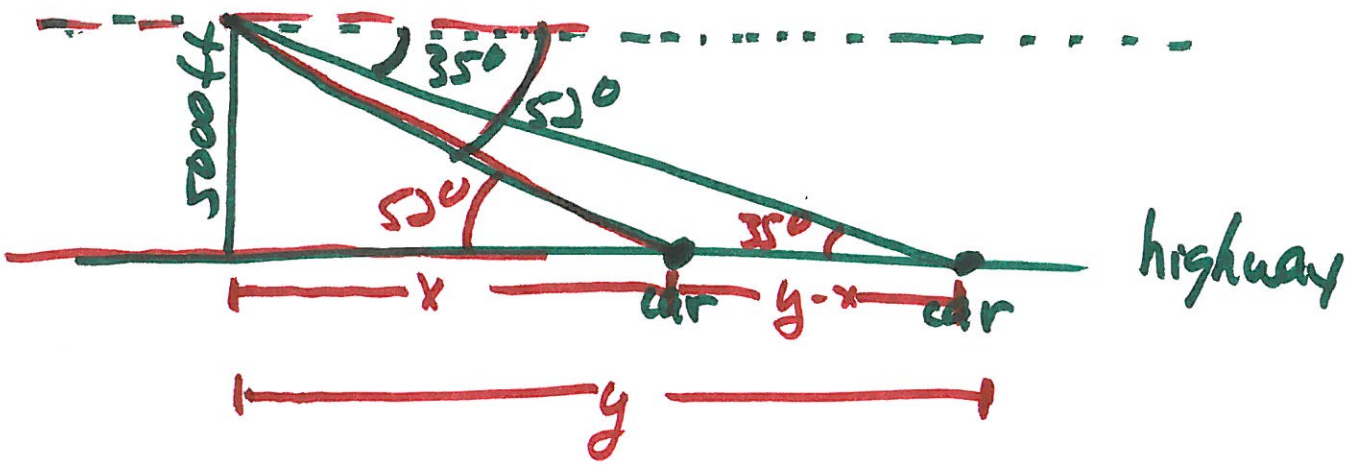
$$\tan 52^\circ = \frac{5000}{x} \rightarrow x = \frac{5000}{\tan 52^\circ}$$

$$\tan 35^\circ = \frac{5000}{y} \rightarrow y = \frac{5000}{\tan 35^\circ}$$

$$x+y = \boxed{\frac{5000}{\tan 52^\circ} + \frac{5000}{\tan 35^\circ}}$$

$$\approx 3906\text{ ft} + 7141\text{ ft} \approx 2\text{ mi.}$$

ex. If both cars in the previous example are on the side of the plane and if the angle of depression to one car is 35° and to the other is 52°, how far apart are the cars?



Find y-x.

$$\tan 52^\circ = \frac{5000}{x} \rightarrow x = \frac{5000}{\tan 52^\circ}$$

$$\tan 35^\circ = \frac{5000}{y} \rightarrow y = \frac{5000}{\tan 35^\circ}$$

$$y - x = \boxed{\frac{5000}{\tan 35^\circ} - \frac{5000}{\tan 52^\circ}} \approx \frac{1}{2} \text{ mi}$$

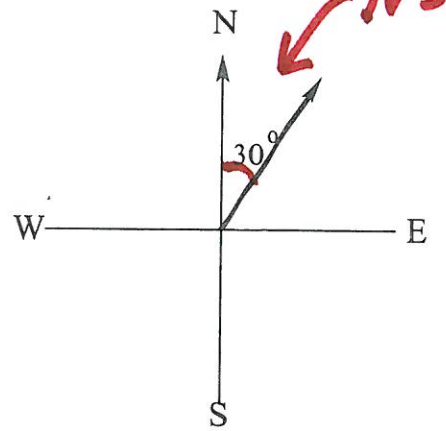
Navigation: Bearing

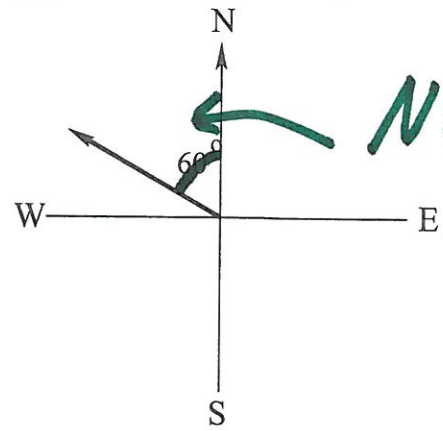
In navigation a direction is often given as a **bearing**, that is, as an acute angle measured from due north or due south.

"30° E of N"

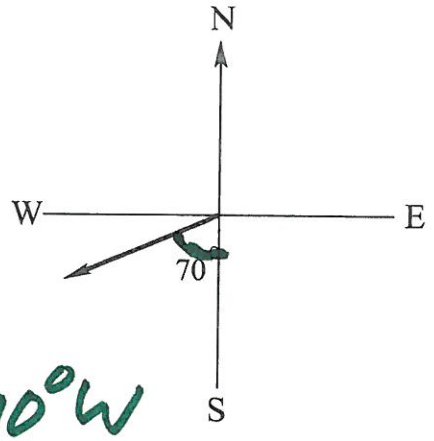
N30°E

"60° W of N"

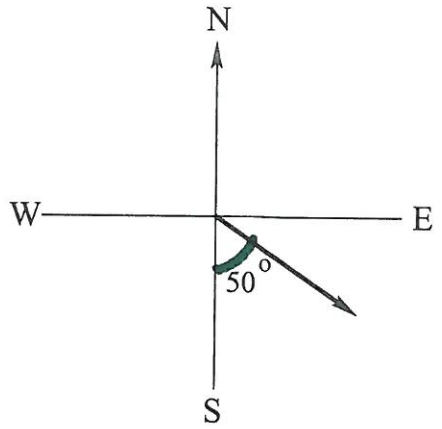




N60°W



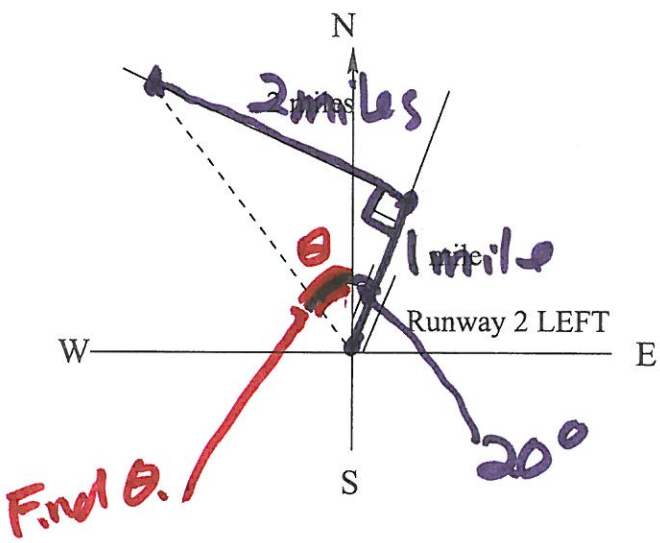
S70°W

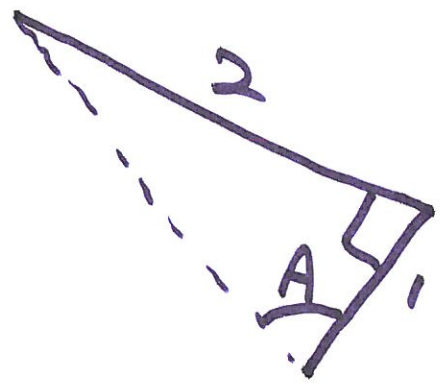


S50°E

"70° W of S"

ex. An aircraft takes off on runway 2 LEFT, which has a bearing of N20°E. After flying for 1 mile, the pilot of the aircraft requests permission to turn 90° and head toward the north west. The request is granted. After the plane goes 2 miles in this direction, what bearing should be the control tower use to locate the aircraft?





$A = \theta + 20°$

$\tan A = \frac{2}{1} \rightarrow A = \arctan 2$

Note: $A = \theta + 20° \rightarrow \theta = A - 20°$

$\theta = \underbrace{\arctan 2}_{} - 20°$

calc. $\theta = 63.4° - 20°$

$\theta = 43.4°$

thus bearing $\boxed{N\ 43.4°\ W}$